Lesson 6.11 — Unit 6 Review and Assessment

Big idea: Unit 6 is exponentials, logs, and how to switch between them. Logs are exponents in disguise; e is the base of continuous growth.

Quick review

Exponential form: $b^y = x$. Logarithmic form: $\log_b(x) = y$. Same statement.
Properties of logs: product, quotient, power, change of base.
Solving: exponential ↔ take log; log ↔ exponentiate.
Real-world: growth/decay $(1 \pm r)^t$ or e^{kt}; log scales for huge dynamic ranges.

Worked example

Worked review. Solve $5^x = 100$. Take log: $x = \log 100 / \log 5 = 2/0.6990 \approx 2.861$.

Warm-ups

1. Evaluate $\log_2(8)$.
2. Evaluate $\ln(e^4)$.
3. Solve $2^x = 32$.

Practice

1. Solve $3^{x+1} = 27$.
2. Solve $\log_5(x) = 3$.
3. $1,000 at 4% continuously compounded for 10 years.

Challenge

1. An isotope has half-life 8 days. How long until 12.5% remains?
2. Solve $\log(x + 1) + \log(x - 2) = 1$.

Answer key — Lesson 6.11

Check your work. If a problem tripped you up, rewatch the step in the video where that concept appears.

Warm-ups

1. 3.
2. 4.
3. $x = 5$.

Practice

1. $x + 1 = 3 \Rightarrow x = 2$.
2. $x = 125$.
3. $1000\,e^{0.4} \approx \$1,491.82$.

Challenge

1. $0.125 = (0.5)^3 \Rightarrow$ 3 half-lives $\Rightarrow$ 24 days.
2. $\log[(x + 1)(x - 2)] = 1 \Rightarrow x^2 - x - 2 = 10 \Rightarrow x^2 - x - 12 = 0 \Rightarrow (x - 4)(x + 3) = 0 \Rightarrow x = 4$ (reject -3 since $x - 2$ must be > 0).